Alibi Shokputov

Urban and Regional Planning and Design
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Education

University of Maryland — College Park, MD
Ph.D. in Urban and Regional Planning and Design — *Expected Spring 2027*
- GPA: 3.83/4.0; Dean's Fellowship; Ann G. Wylie Dissertation Fellowship
- Dissertation: *The Unequal Price of Proximity: Telework, Commute Frequency, and Place-Tied Workers in U.S. Cities*
- Advisor: Casey Dawkins

Ulsan National Institute of Science and Technology (UNIST) — Ulsan, South Korea
M.Sc. in Urban Infrastructure Engineering — *2018 – 2020*
- GPA: 4.08/4.3; Full Graduate Scholarship for two years
- Thesis: *Impact of Regional Transportation Investments on Housing Values in Central Florida*

Ulsan National Institute of Science and Technology (UNIST) — Ulsan, South Korea
B.Sc. in Urban Infrastructure Engineering & Business Management (Minor) — *2014 – 2018*
- GPA: 3.86/4.3; Full Scholarship for four years; Graduated with Honors: *Magna Cum Laude*

Research and Professional Appointments

Department of Civil and Environmental Engineering, University of Maryland
Research Assistant — Post-Disaster Housing Recovery and Regulatory Policy (HUD-funded) — *Oct 2026 – Present*
- Develop reproducible Python workflows to collect and standardize zoning, building-code, permitting, and tax ordinances through APIs and web scraping.
- Implement LLM-assisted extraction with human validation to classify policy changes and document their provisions, rationales, geographic scope, and effective dates.
- Compile a documented policy database and descriptive analyses to support the team's panel construction and causal research on rebuilding speed and housing supply.

Real Estate Development, University of Maryland
Research Assistant — Rental Affordability in Greener Neighborhoods — *Summer 2026 – Present*
- Analyzed disparities in tree canopy, heat, impervious surfaces, and flood exposure across NOAH (naturally occurring affordable housing), LIHTC, and market-rate housing in the Washington, D.C.–Baltimore corridor.
- Developed the *Alpha Affordability Gradient*, an average median income-based metric to define NOAH and evaluate alternative affordability thresholds.
- Built a geospatial data pipeline and applied spatial econometric methods to identify within- and between-neighborhood disparities, support policy recommendations, and prepare a manuscript for the *Journal of Sustainable Real Estate*.

Data Science for Social Good Fellowship, Johns Hopkins University — Baltimore, MD
Summer Fellow — Baton Rouge Blight Remediation — *Summer 2026*
- Selected for a competitive 10-week fellowship with Johns Hopkins' GovEx and Baton Rouge's Department of Development to proactively address urban blight before resident complaints.
- Developed a machine-learning pipeline combining spatial contagion and temporal signals to predict at-risk properties early, enabling inspectors to intervene first.
- Designed equitable interventions by directing distressed homeowners to assistance before enforcement, reserving action for absentee and investor neglect, and auditing predictions based on risk rather than complaints.

Lab @ DDOT (District Department of Transportation) — Washington, D.C.
Research Assistant — Bike Ridership Prediction in Washington, D.C. — *Summer 2025*
- Developed a machine-learning framework to predict citywide bike ridership from sparse, uneven data, revealing unmet demand for safe cycling infrastructure not previously seen by the agency.
- Collaborated with data and social scientists at The Lab @ DDOT, housed within DDOT's Innovation Branch, to advance agency and community priorities through evidence-based transportation research.

National Center for Smart Growth Research, University of Maryland
Research Assistant *Jan 2022 – May 2025*

2022–2024 Annual Maryland Commuter Survey
- Produced three annual reports for MDOT by analyzing 2,621 worker responses, delivering technical analyses, visualizations, policy insights, and briefings for state officials.
- Built reusable infrastructure for survey design, data cleaning, weighting, commute mapping, and documentation, enabling efficient and reproducible future survey waves.

Evaluating the Maryland Commuter Tax Credit Program Jan 2024 – Aug 2024
- Lead-authored an MDOT evaluation documenting severe underuse, with employers claiming less than $160,000 of the $1 million authorized annually, even after eligibility expanded to eight commuter benefits in 2022.
- Evaluated program records and supervised stakeholder interviews and focus groups, identifying administrative burden, uncertain employer ROI, and limited transit access as key adoption barriers.
- Proposed employee-level filing, broader eligibility, and simplified administration and outreach to expand access for low-income and nontraditional workers.

Multi-Criteria Decision-Making Framework for Public Microtransit Jan 2022 – Aug 2023
- Developed a replicable framework combining the Microtransit Propensity Index with a new accessibility score to rank census blocks by microtransit suitability and unmet access to jobs and essential destinations.
- Synthesized evidence from 33 successful U.S. microtransit programs and applied the framework to four candidate areas selected by Prince George's County.
- Presented recommendations to county officials and co-authored a peer-reviewed article in *Transportation Research Record*.

District Department of Transportation (DDOT) Washington, D.C.
Transit Operations Data Analyst Intern *Summer 2023*
- Built a real-time Tableau dashboard tracking bus and operator assignments, route-level fleet compliance against scheduled service, vehicle status histories, and EV charging availability and battery levels.
- Standardized data-entry protocols for RATP Dev personnel, improving the consistency and reliability of operational reporting and dashboard data.

Selected Publications

1. Erdoğan, S., Nohekhan, A., **Shokputov, A.**, & Sadabadi, K. F. (2023). Multi-Criteria Decision-Making Framework for Determining Public Microtransit Service Zones. *Transportation Research Record*. DOI.

Manuscripts in Preparation

1. **Shokputov, A.** Who Teleworks and Who Wants To? Preferences, Feasibility, and Telework Misalignment Among Workers in Teleworkable Jobs. *In preparation.*
2. **Shokputov, A.** The Rent-Wage Scissors: Telework and the Divergence of Housing-Cost and Earnings Gradients for Place-Tied Workers in U.S. Metropolitan Areas. *In preparation.*
3. Park, J., & **Shokputov, A.** Property-Level Affordability and Environmental Exposure in Multifamily Housing: Canopy, Heat, and Flood in the Washington–Baltimore Corridor. *In preparation.*
4. **Shokputov, A.**, & Dawkins, C. Telework Intensity and Realized Home-Work Separation: Disentangling Sorting from Behavioral Change. *In preparation.*
5. **Shokputov, A.** When Hybrid Work Does Not Reduce Commuting: Stated Arrangements, Behavioral Discordance, and Weekly Commute Exposure. *In preparation.*

Grants and Awards

MPower Early Scholars Investment Fund: GA stipend and tuition support, University of Maryland — Spring 2027
Data Science for Social Good Fellowship: $14,500, Johns Hopkins University — Summer 2026
Ann G. Wylie Dissertation Fellowship: $15,000, University of Maryland — Spring 2026
Dean's Fellowship: $20,000, University of Maryland — 2021–2022
Best Conference Paper Award: Korea Planners Association — Fall 2018

Teaching Appointments

Advanced Research Methods for Urban Analysis (Graduate Level) Winter 2027 (forthcoming)
Instructor and Course Developer, University of Maryland

- Conceived and developed a new interdisciplinary graduate elective from the ground up for Winter 2027.
- Designed advanced training in methodological judgment, emphasizing spatial scale, measure construction, and model selection with complex real-world data.
- The course will guide students through an independent, portfolio-ready study in R, translating results into defensible maps, figures, and policy conclusions.

URSP601 Quantitative Research Methods (Graduate Level) Fall 2024, Fall 2025
Course Instructor, University of Maryland

- Overhauled the core URSP601 syllabus as sole instructor for two graduate cohorts, modernizing foundational methods training through R.
- Created hands-on coursework using administrative and spatial data to teach cleaning, visualization, exploratory analysis, and reproducible workflows.
- Taught statistical inference for urban policy applications; mentored students across varying skill levels and received strong evaluations.

Conference Papers

1. **Shokputov, A.** (2026). Hybrid Without the Telework: Discordance Between Stated Arrangements and Commute Behavior. *ACSP*, Pittsburgh, PA.
2. Park, J., & **Shokputov, A.** (2026). Property-Level Affordability and Environmental Exposure in Multifamily Housing: Canopy, Heat, and Flood in the Washington–Baltimore Corridor. *ACSP*, Pittsburgh, PA.
3. **Shokputov, A.** (2026). The Rent-Wage Scissors? Telework and Divergent Accessibility Gradients for Place-Tied Workers. *Data-Intensive Research Conference (IPUMS)*, Minneapolis, MN.
4. **Shokputov, A.**, Harvey, C. (2025). Beyond Feasibility: Telework, Commuting, and Equity in Post-Pandemic Maryland. *ACSP*, Minneapolis, MN.
5. **Shokputov, A.**, Harvey, C. (2024). Uncovering the Latent Demand for Telecommuting in Maryland. *ACSP*, Seattle, WA.
6. **Shokputov, A.**, Harvey, C. (2023). Maryland's Remote Work Landscape: Commuting Behaviors and Potential VMT Reductions. *ACSP*, Chicago, IL.
7. **Shokputov, A.**, Iseki, H. (2023). Key Factors Driving Plug-in Electric Vehicle Adoption. *Transportation Research Forum*, Chicago, IL.
8. Zou, Z., **Shokputov, A.**, & Erdoğan, S. (2022). Commuting Trends during the COVID-19 Pandemic: Evidence from Maryland. *TRB Annual Meeting*, Washington, D.C.

Technical Toolkit

- **Programming Languages**: Python, R
- **Machine Learning**: scikit-learn, XGBoost
- **GIS and Spatial Analysis**: ArcGIS Pro, QGIS, PostGIS, GDAL, rasterio, GeoPandas, OSMnx, sf, tidycensus
- **Visualization**: Tableau, Shiny
- **Data and Databases**: SQL, PostgreSQL, BigQuery, SQLAlchemy
- **Cloud and Workflow**: Google Cloud, Docker, Git
- **Reporting and Survey**: LaTeX, Qualtrics